

Supervised Learning Algorithms: Linear Regression, Logistic Regression, and Decision Trees

Introduction

Supervised learning is a machine learning approach in which a model learns from labeled training data. The objective is to learn a mapping between input features and target outputs so that predictions can be made on unseen data.

Supervised learning problems are broadly categorized into:

1. Regression
2. Predicts continuous numerical values.
3. Example: House price prediction.
4. Classification
5. Predicts categorical labels.
6. Example: Email spam detection.

This report explores three commonly used supervised learning algorithms: Linear Regression, Logistic Regression, and Decision Trees.

1. Linear Regression

Definition

Linear Regression is a regression algorithm used to predict continuous values by modeling a linear relationship between independent variables and a dependent variable.

Working Principle

The algorithm fits a straight line: $Y =$

$mX + c$ where:

- Y = Target variable
- X = Feature
- m = Slope
- c = Intercept

The model minimizes the sum of squared errors between actual and predicted values.

Assumptions

- Linear relationship between variables

- Independence of observations
- Homoscedasticity
- No multicollinearity
- Normally distributed residuals

Applications

- House price prediction
 - Sales forecasting
 - Demand estimation
 - Stock trend analysis
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2. Logistic Regression

Definition

Logistic Regression is a classification algorithm used for predicting categorical outcomes.

Working Principle

It uses the sigmoid function:

$$P(Y=1) = 1 / (1 + e^{(-z)})$$

The output represents probability values between 0 and 1.

Assumptions

- Independent observations
- Minimal multicollinearity
- Linear relationship between features and log odds

Applications

- Disease prediction
 - Spam detection
 - Customer churn prediction
 - Fraud detection
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3. Decision Tree

Definition

Decision Trees are supervised learning algorithms that make predictions by splitting data into branches based on feature values.

Working Principle

The algorithm repeatedly selects the feature that best separates the data using criteria such as:

- Gini Index
- Entropy
- Information Gain

The final prediction is made at leaf nodes.

Assumptions

- No strict assumptions about data distribution
- Can handle both numerical and categorical features

Applications

- Medical diagnosis
- Credit risk analysis
- Customer segmentation
- Recommendation systems

Conclusion

Linear Regression is suitable for continuous prediction tasks, Logistic Regression is effective for classification problems, and Decision Trees provide flexible solutions for both regression and classification. Understanding these algorithms forms the foundation for advanced machine learning techniques such as Random Forests, Gradient Boosting, and Ensemble Learning.